

Quiz 2 solutions

OPTIMIZERS 1

CS312

Tatsu H

Quiz instructions..

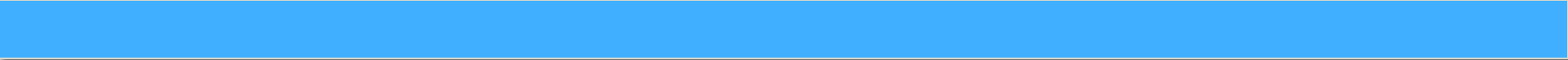
Please take your quiz from the front

We will start at 1:35.

The quiz is 40 minutes, and ends at 2:15

The answer sheet is the *last page*. Please write your Sunet ID (not your ID #)

There are 3 citation/reasoning fields for you to fill in for partial credit.



Question 1a:

Change from ‘1000 different blocks’ to ‘1000 shared param block’

This question changes. Depth $L = 1000$, so $r = 500$ and c_L is 1, .0447 and .002 for A, B and C. The 1,000 blocks are **one shared block**: a single block, initialized like one block of the stress test, applied 1,000 times with shared weights. It takes its prescription's block LR and ϵ .

(a) Residual stream at initialization (12 points, 4 per row). Before any update, take the RMS of the final residual stream (the input to the final RMSNorm, pooled over the 8 fixed probe sequences). Let

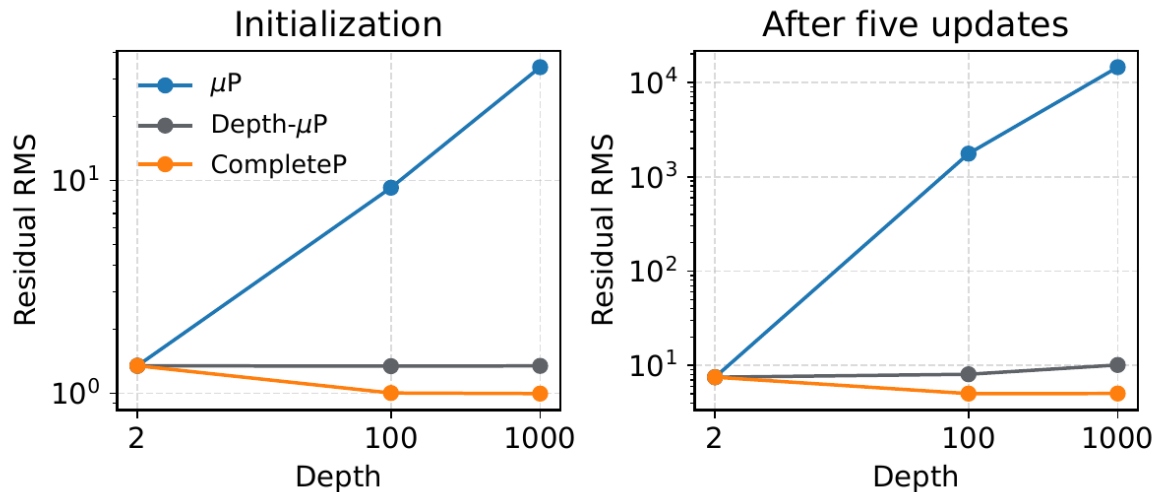
$$\rho = \frac{\text{this RMS in the shared-block model}}{\text{this RMS in the stress-test model with 1,000 independent blocks'}}$$

with the same prescription and seed. Which interval contains ρ ?

Prescription	Option 1	Option 2	Option 3	Option 4
A. Standard μP	$\cdot (0, 10)$	$\cdot [10, 100)$	$\cdot [100, 1000)$	$\cdot [1000, \infty)$
B. Depth- μP	$\cdot (0, 10)$	$\cdot [10, 100)$	$\cdot [100, 1000)$	$\cdot [1000, \infty)$
C. CompleteP	$\cdot (0, 10)$	$\cdot [10, 100)$	$\cdot [100, 1000)$	$\cdot [1000, \infty)$

Evidence from A2

We'll show some data from our playtest first..

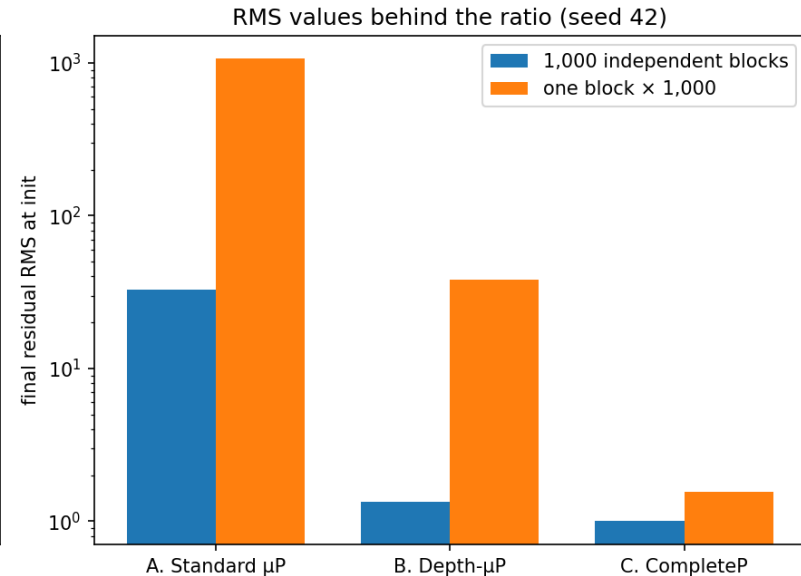
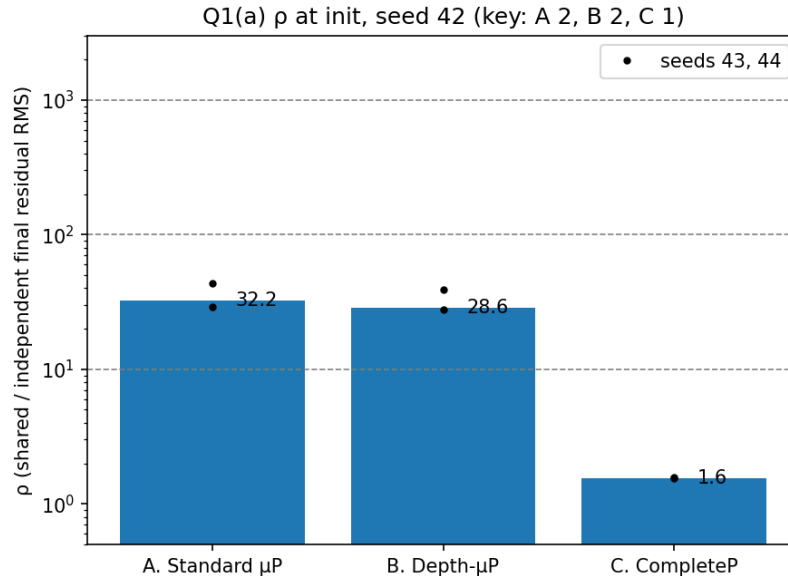


In the assignment (4.1) – depth μP worked “better” and completeP overcorrected.

Answer

Correct options: 2, 2, 1

Prescription	Option 1	Option 2	Option 3	Option 4
A. Standard μP	$\cdot (0, 10)$	$\cdot [10, 100)$	$\cdot [100, 1000)$	$\cdot [1000, \infty)$
B. Depth- μP	$\cdot (0, 10)$	$\cdot [10, 100)$	$\cdot [100, 1000)$	$\cdot [1000, \infty)$
C. CompleteP	$\cdot (0, 10)$	$\cdot [10, 100)$	$\cdot [100, 1000)$	$\cdot [1000, \infty)$



Ratios similar for standard and dept μP but not for completeP. Why?

Rationale

Standard, Depth, CompleteP all just shift the residual contribution

The *ratio* at init should thus be constant on the residual sums.

Prescription	c_L	RMS, independent	RMS, shared	ρ	Branch sum, independent $\rightarrow$ shared	Branch-sum ratio
A. Standard μP	1	33	1,062	32.2	33 $\rightarrow$ 1,062	$\approx 32\times$
B. Depth- μP	$r^{-1/2} \approx .045$	1.34	38.4	28.6	0.89 $\rightarrow$ 38.4	$\approx 43\times$
C. CompleteP	$r^{-1} = .002$	1.00	1.56	1.56	0.036 $\rightarrow$ 1.20	$\approx 34\times$

But! Notice how RMS is not just about the residuals sums. Embeddings are different
(think about the extreme of zeroing out the residual contribution)

Question 1b

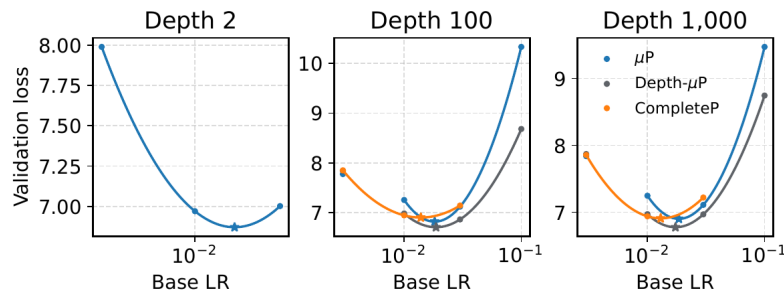
(b) Loss after five updates (8 points). For each prescription, take the shared-block model's lowest validation loss after five updates over base LR's $\{.003, .01, .03\}$. Which prescription attains the lowest of these three losses?

- 1. A. Standard μP · 2. B. Depth- μP · 3. C. CompleteP · 4. All three within .005

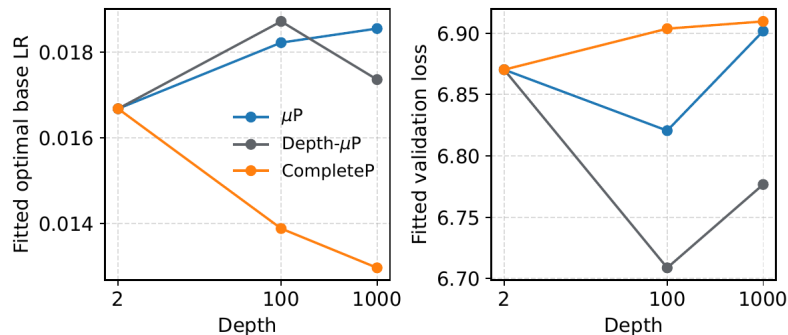
Five-step stability version

From A2

Data from A2: Depth muP is the best for large depth



Original five-update loss sweeps. Dots are measured losses; stars are minima of local quadratics in log LR. All three prescriptions share the depth-two reference.

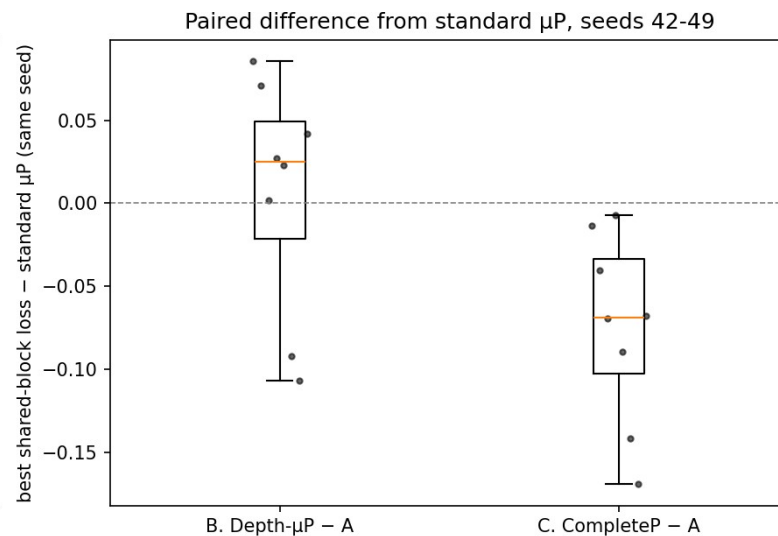
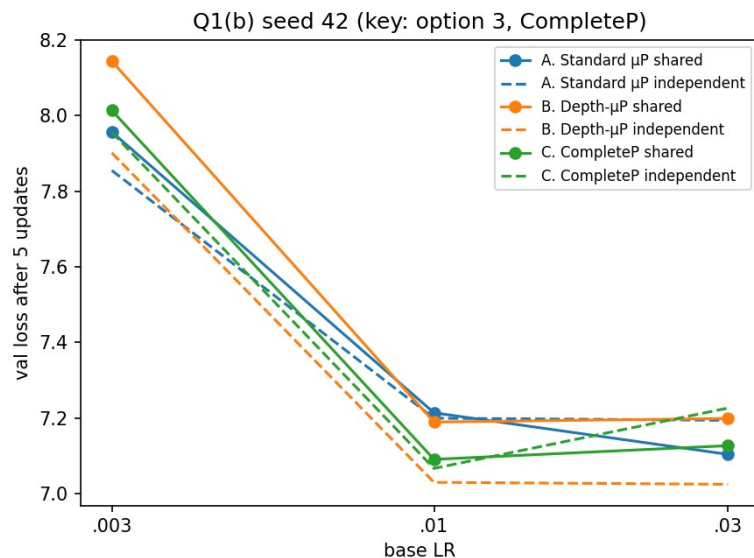


(b) **Loss after five updates (8 points).** For each prescription, take the shared-block model's lowest validation loss after five updates over base LR_s $\{.003, .01, .03\}$. Which prescription attains the lowest of these three losses?

Q1b

1. A. Standard μP 2. B. Depth- μP 3. C. CompleteP 4. All three within .005

Best: CompleteP. How could we have known?



Rationale: complete P targets full alignment. This is exactly the ‘block copying’ case.

Q2a

This question changes. The LR schedule is **cosine decay to zero** instead of linear decay (same 1% warmup), at **1.2288B tokens**, on the same 3×3 LR-WD grid. With cosine decay, the grid's best cell is peak LR .003, WD .2 ($\eta\lambda = 6 \times 10^{-4}$), with loss 2.8354. Configurations A–F are six other cells of this grid:

Configuration	Peak LR η	WD λ	$\eta\lambda (\times 10^{-4})$
A	.006	.1	6
B	.0015	.4	6
C	.006	.4	24
D	.0015	.1	1.5
E	.006	.2	12
F	.003	.4	12

Let $L_A, \dots, L_F$ denote the losses of configurations A–F.

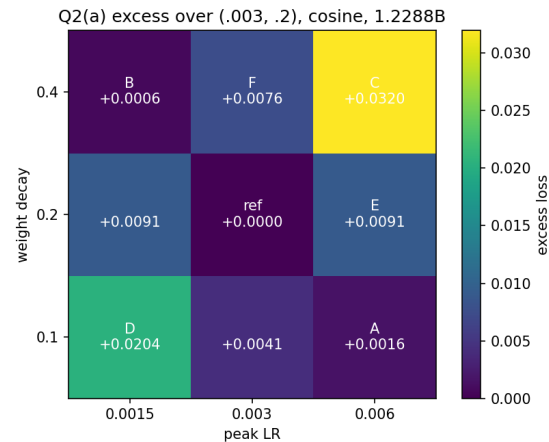
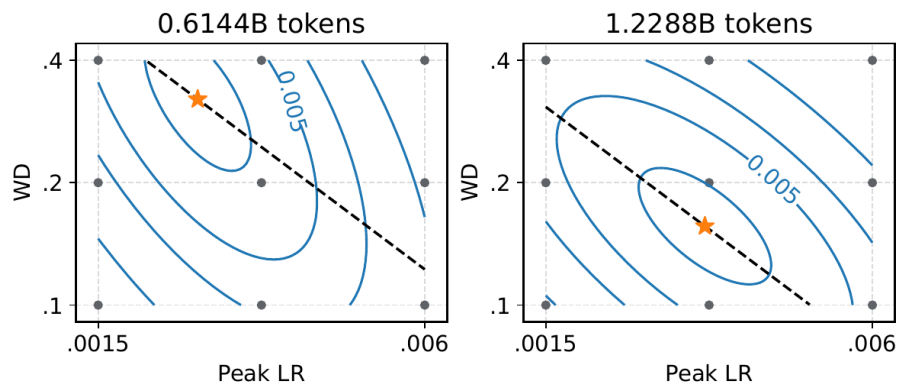
(a) Which loss ranking do you predict? (10 points) Choose the option closest to the measured ranking (fewest pairs in the wrong order).

1. $L_A < L_B < L_F < L_E < L_D < L_C$
2. $L_D < L_B < L_F < L_E < L_A < L_C$
3. $L_F < L_E < L_A < L_B < L_C < L_D$
4. $L_C < L_E < L_F < L_A < L_B < L_D$

A2 information

Recall your joint WD/LR plots

Rough rule of thumb – deviation from optimal is ‘best’ if you keep the WD*LR sum

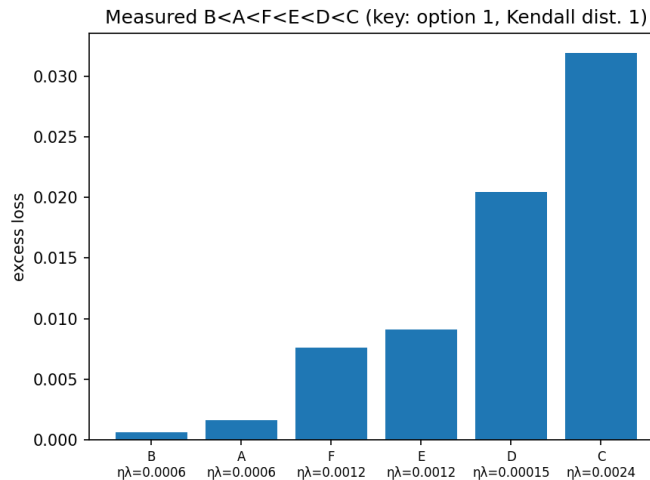
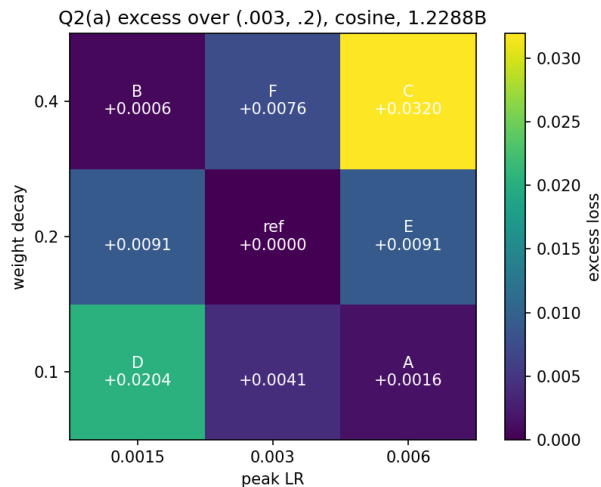


(a) Which loss ranking do you predict? (10 points) Choose the option closest to the measured ranking (fewest pairs in the wrong order).

1. $L_A < L_B < L_F < L_E < L_D < L_C$
2. $L_D < L_B < L_F < L_E < L_A < L_C$
3. $L_F < L_E < L_A < L_B < L_C < L_D$
4. $L_C < L_E < L_F < L_A < L_B < L_D$

Q2 Answer

Read straight off of the assignment – A B F E D C is the closest order.



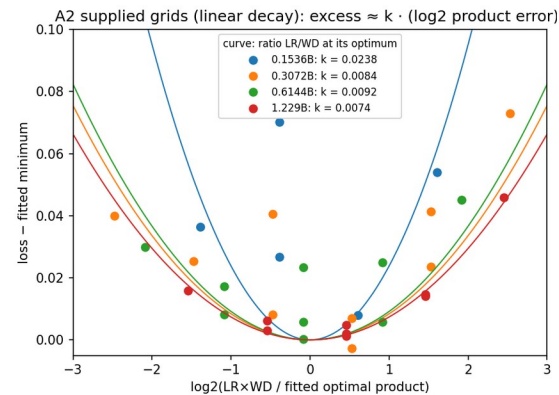
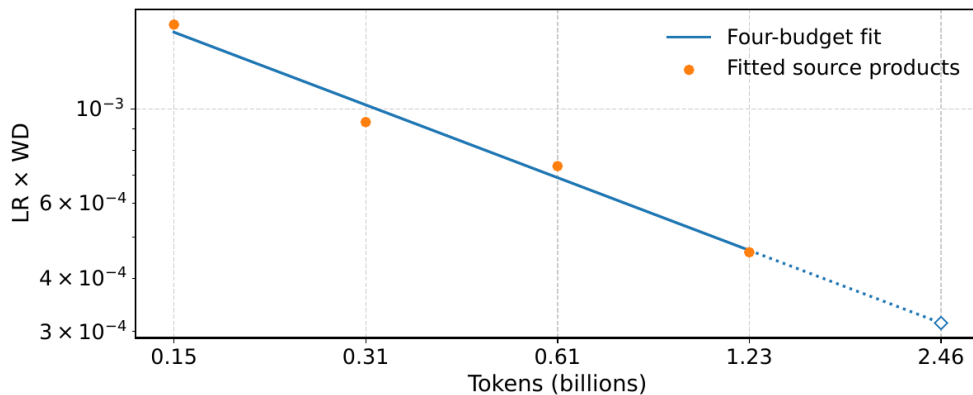
Q2b

(b) (10 points) Which interval contains configuration C's excess loss, $L_C - 2.8354$?

- 1. $[0, .005)$ · 2. $[.005, .015)$ · 3. $[.015, .025)$ · 4. $[.025, \infty)$

This is the numerical prediction followup

Q2b assignment info

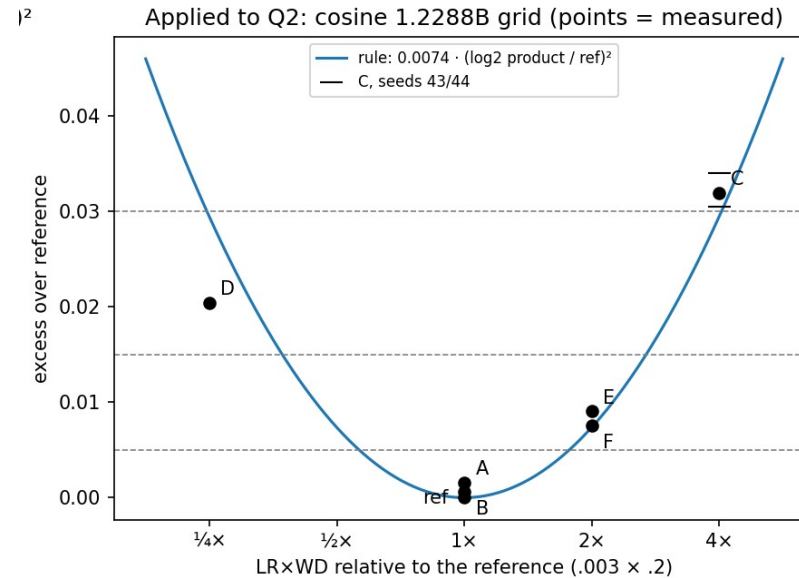


Roughly, quadratic penalty to being “off” on the product, with $0.007 \cdot \log \text{product}^2$

(b) (10 points) Which interval contains configuration C's excess loss, $L_C - 2.8354$?

1. $[0, .005)$ 2. $[\cdot005, \cdot015)$ 3. $[\cdot015, \cdot025)$ 4. $[\cdot025, \infty)$

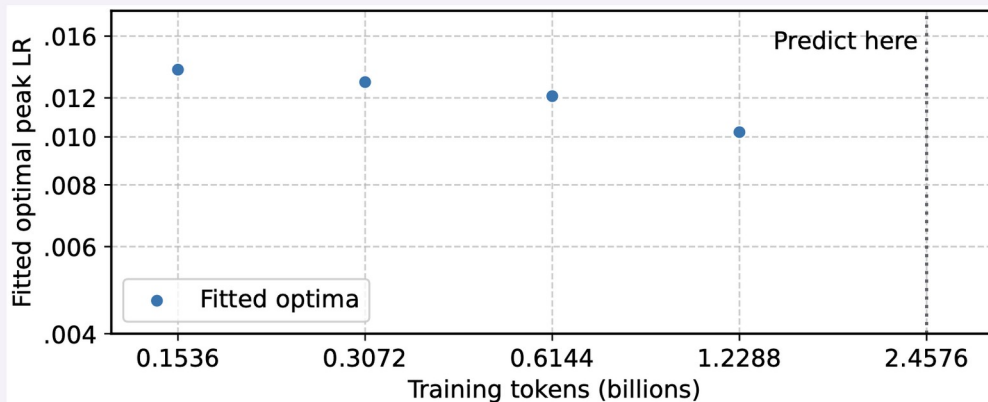
Answer / rationale:



This “rule of thumb” give us ~ 0.03 , which resolves to option 4 ($\cdot = 0.25$)

Q3

From the assignment (reference only). Recall our study of Hyperball. We ran scaling ladders, with Hyperball for every linear-layer weight matrix (including the readout), and Adam on embeddings and normalization gains at peak LR $.1041 \eta$, where η is the Hyperball peak LR; for this problem, we disabled weight decay. For reference, we provide the fitted optima of three supplied budgets and of the 1.2288B target on log-log axes.

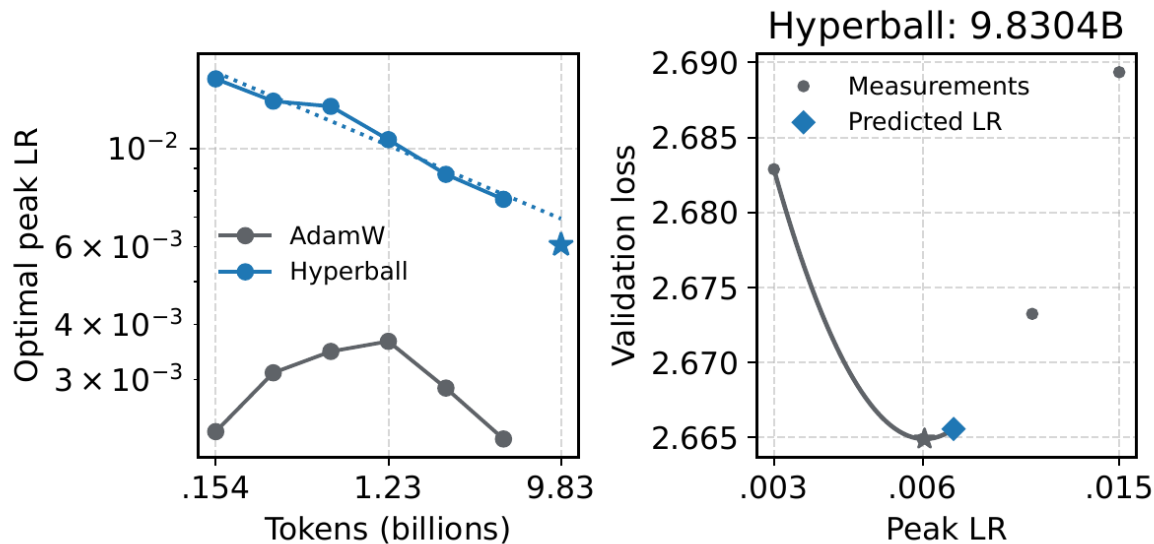


This question changes. The token budget is **2.4576B tokens** (the dotted line), with peak LR $\{.006, .01, .015\}$.

Which interval contains the fitted optimum at 2.4576B tokens? (20 points)

- 1. $[\cdot0075, \infty)$
- 2. $[\cdot0055, \cdot0075)$
- 3. $[\cdot004, \cdot0055)$
- 4. $(0, \cdot004)$

A2 data

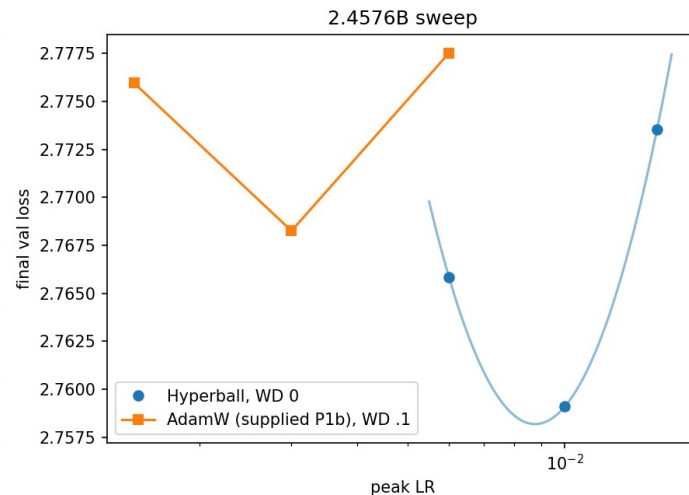
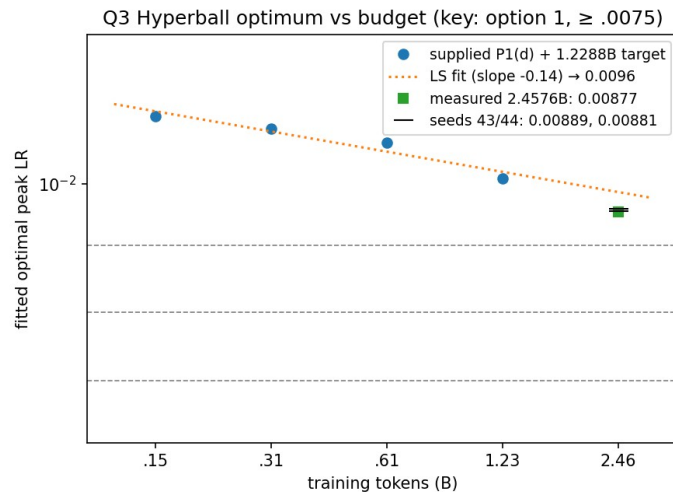


This was a slightly scaled up version of the assignment sweep.

Main takeaway – you can (roughly) extend the line on hyperball

Q3 answer

1. $[\cdot0075, \infty)$ 2. $[\cdot0055, \cdot0075)$ 3. $[\cdot004, \cdot0055)$ 4. $(0, \cdot004)$



Just follow the trendline to get the answer (~ 0.008)

Q4

From the assignment (reference only). Recall that we ran experiments varying momentum at 614.4M tokens. We build upon that set of experiments, keeping $\beta_2 = 0.95$ fixed, and varying only β_1 .

This question changes. We only consider two different batch size / learning rate / weight decay combinations, and β_1 takes three values: **A:** $\beta_1 = 0$, **B:** $\beta_1 = .9$, **C:** $\beta_1 = .98$.

Batch size B	Peak LR	WD	Loss at $\beta_1 = .9$
8	.00075	.1	2.936603
256	.0015	1.944	2.968249

At each batch size, let L_A, L_B, L_C be the losses with $\beta_1 = 0, .9, .98$.

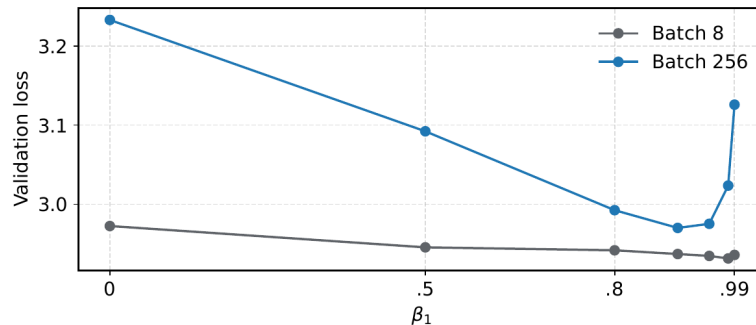
(a) Rankings (5 points per batch size).

Option 1: $L_C < L_B < L_A$ **Option 2:** $L_A < L_B < L_C$

Option 3: $L_A < L_C < L_B$ **Option 4:** $L_B < L_C < L_A$

Choose one option for $B = 8$ and one for $B = 256$.

A2 data



Both complete momentum curves share an axis. LR and WD stay fixed within each batch, and $\beta_2 = .95$.

β_1	Batch 8 loss	Batch 256 loss
0	2.972512	3.232685
0.5	2.945586	3.092095
0.8	2.941873	2.992511
0.9	2.937171	2.970108
0.95	2.934768	2.975301
0.98	2.931576	3.023549
0.99	2.936018	3.125746

Very characteristic momentum shape – flat in batch 8, sharp in batch 256

Q4 answer

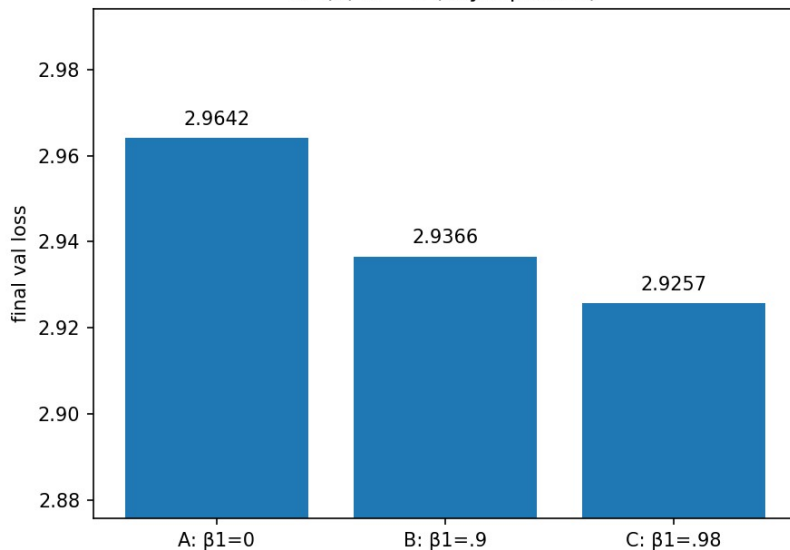
(a) Rankings (5 points per batch size).

Option 1: $L_C < L_B < L_A$ Option 2: $L_A < L_B < L_C$

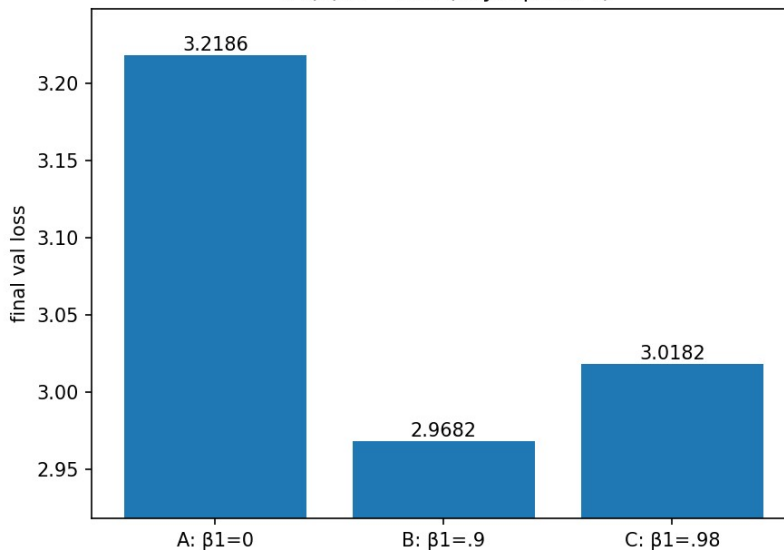
Option 3: $L_A < L_C < L_B$ Option 4: $L_B < L_C < L_A$

Choose one option for $B = 8$ and one for $B = 256$.

Q4(a) $B = 8$ (key: option 1)



Q4(a) $B = 256$ (key: option 4)



Rankings: option 1 (CBA) and option 4 (BCA)

This is basically ‘read off of’ the momentum chart.

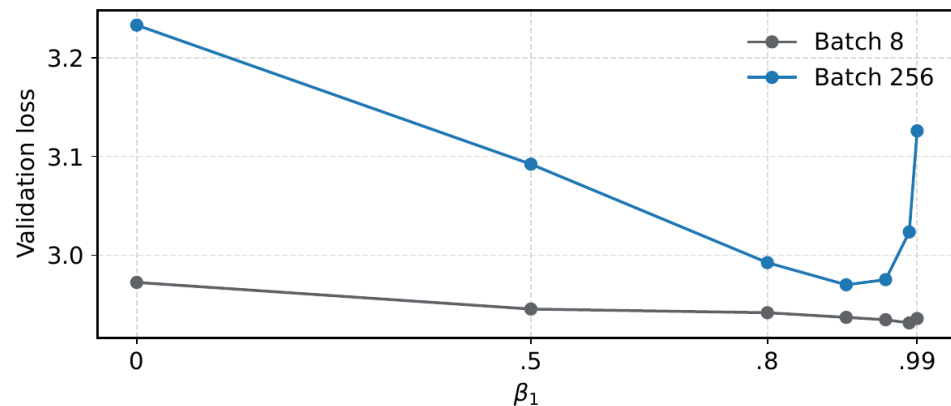
The only / main trickiness is B/C on Batch=8, but knowing the “default” momentum would help.

Part b

(b) Cost of removing momentum (5 points per batch size). Which interval contains $L_A - L_B$?

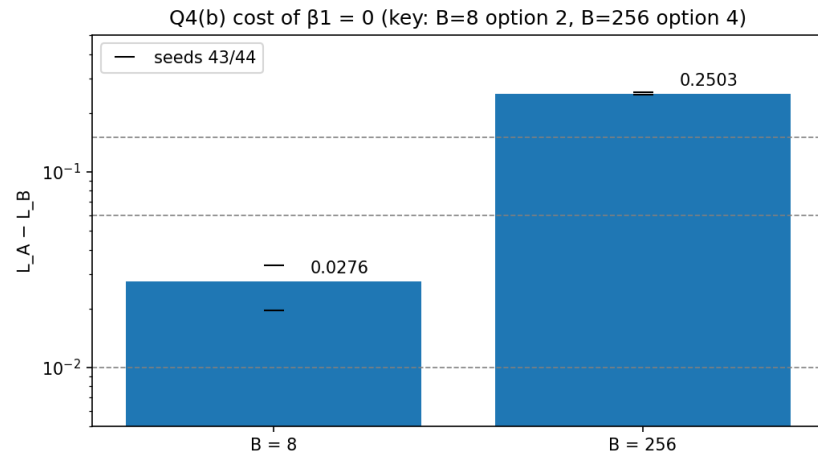
B	Option 1	Option 2	Option 3	Option 4
8	$\cdot (-\infty, .01)$	$\cdot [.01, .06)$	$\cdot [.06, .15)$	$\cdot [.15, \infty)$
256	$\cdot (-\infty, .01)$	$\cdot [.01, .06)$	$\cdot [.06, .15)$	$\cdot [.15, \infty)$

Part b answer



(b) Cost of removing momentum (5 points per batch size). Which interval contains $L_A - L_B$?

B	Option 1	Option 2	Option 3	Option 4
8	$(-\infty, .01)$	$[\cdot01, \cdot06)$	$[\cdot06, \cdot15)$	$[\cdot15, \infty)$
256	$(-\infty, .01)$	$[\cdot01, \cdot06)$	$[\cdot06, \cdot15)$	$[\cdot15, \infty)$



Should know that the cost is low ish for 8, high for 256. Options 2 and 4

Q5

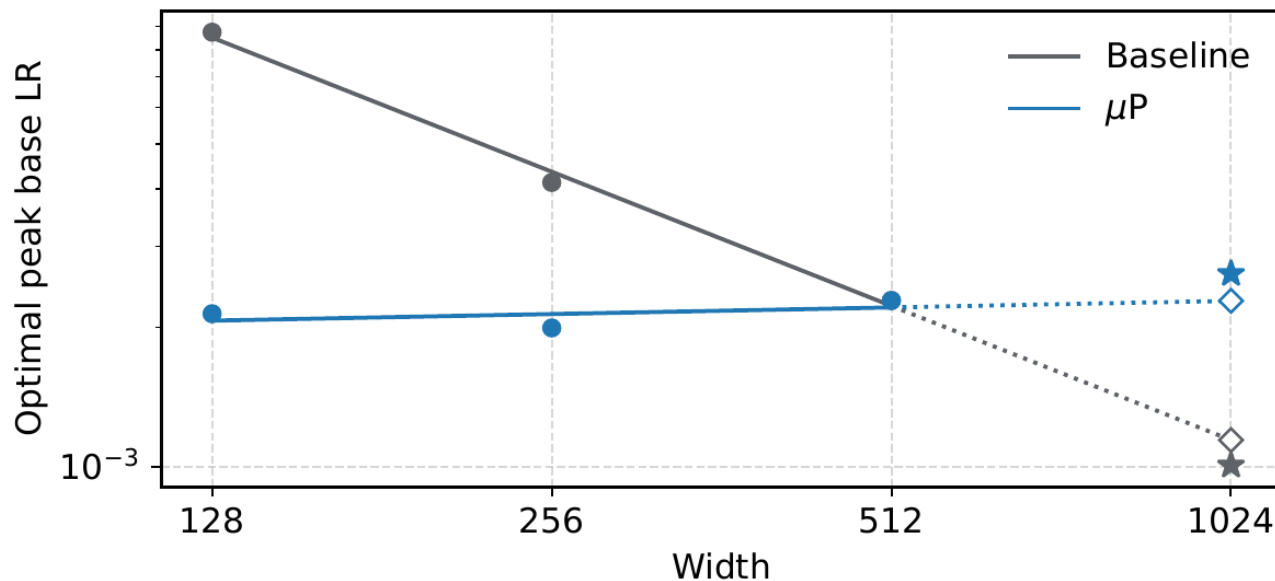
This question changes. A third prescription, **almost- μ P**: μ P's embedding initialization (G/n_0) and hidden-matrix LR (η/m), but the course baseline's readout initialization ($G/\sqrt{n}$) and readout multiplier (1). It is trained at width 1024 ($m = 2$) with peak LRs $\{.00075, .0015, .003, .006, .012\}$.

(a) (10 points) Let $\eta^*(1024)$ be almost- μ P's fitted optimum at width 1024 and $\eta^*(512) = .002378$. Which interval contains $\eta^*(1024)/\eta^*(512)$?

- 1. $(0, .5)$ · 2. $[.5, .8)$ · 3. $[.8, 1.25)$ · 4. $[1.25, \infty)$

Evidence from assignment

Linear LR decay on baseline, flat on μP .



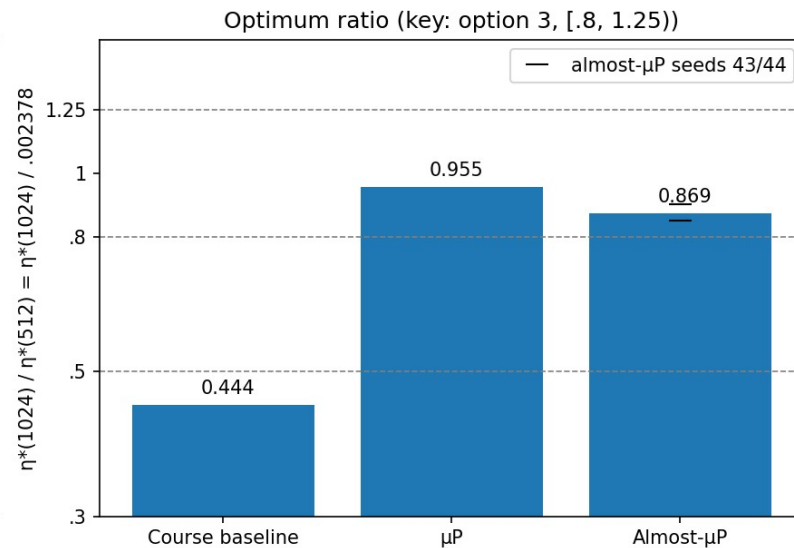
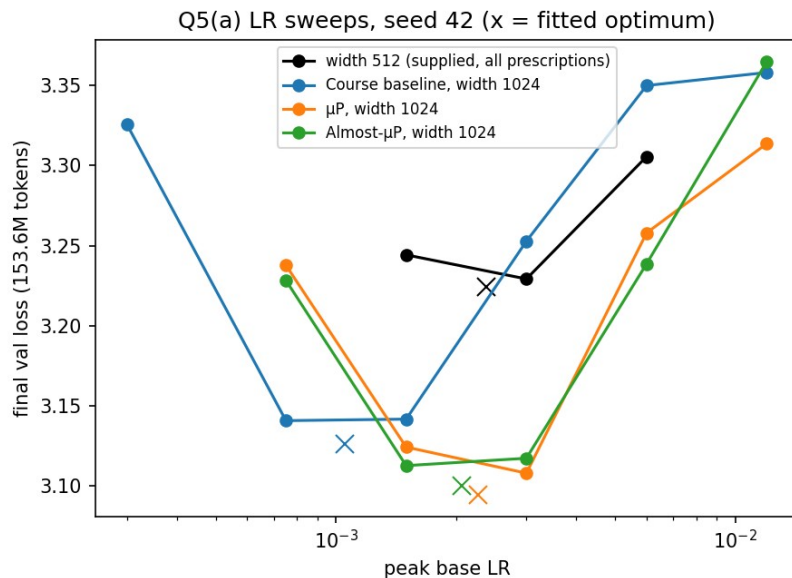
(a) (10 points) Let $\eta^*(1024)$ be almost- μ P's fitted optimum at width 1024 and $\eta^*(512) = .002378$. Which interval contains $\eta^*(1024)/\eta^*(512)$?

1. $(0, .5)$ 2. $[\cdot 5, .8)$ 3. $[\cdot 8, 1.25)$ 4. $[1.25, \infty)$

Answer/rationale

Almost MuP falls in the middle, a bit closer to μ P than baseline.

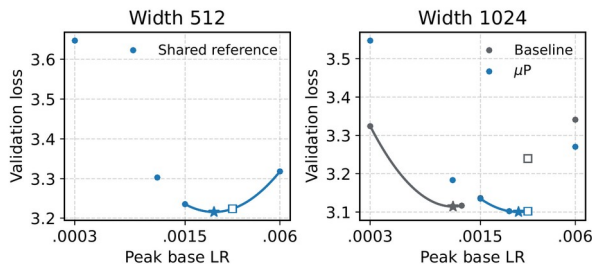
The readout has a smaller contribution



Part b

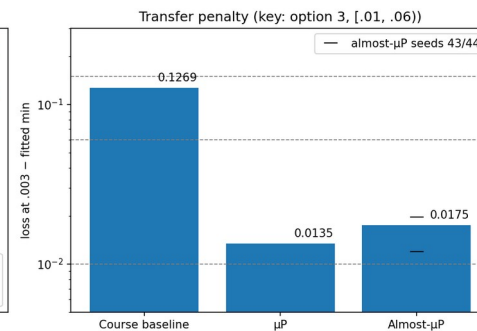
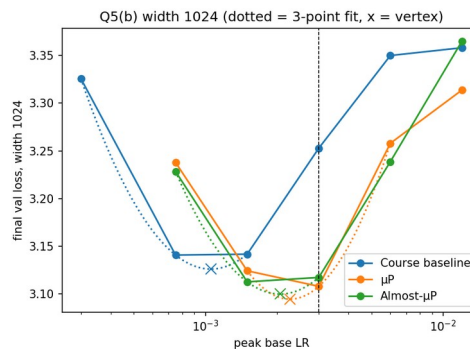
(b) (10 points) At width 1024, which interval contains almost- μP 's transfer penalty for the transferred peak LR .003?

1. $[-.15, \infty)$ 2. $[-.06, .15)$ 3. $[-.01, .06)$ 4. $[0, .01)$



Circles are measurements, stars are local fitted minima, and open squares mark direct transfer of peak base LR .003.

Recipe	Choice	LR	Measured loss	Grid gap
Baseline	Power law	0.001141	3.117064	+0.002288
Baseline	Direct LR	0.003000	3.240047	+0.125271
Baseline	Grid best	0.001000	3.114776	+0.000000
μP	Power law	0.002286	3.102650	-0.000029
μP	Direct LR	0.003000	3.102679	+0.000000
μP	Grid best	0.003000	3.102679	+0.000000



From A2 – we know the transfer penalty is much lower (near 0 in A2).

Our measured data is a bit higher due to suboptimality of our fits, but similar $[0.01, 0.06)$

Q6

From the assignment (reference only).

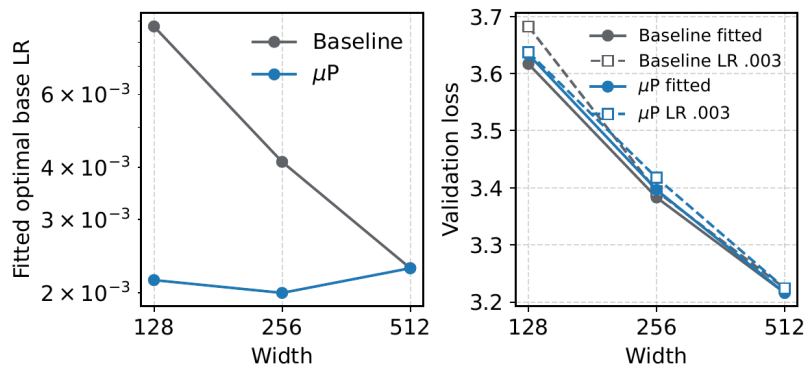
- **Hyperball:** Recall that in Hyperball, every linear-layer weight matrix, including the readout, takes an Adam-direction step and is projected back to its initial Frobenius norm; embeddings and normalization gains use Adam at peak LR $.1041\eta$; for this problem, we disabled weight decay.
- **Course-baseline width sweep:** From our assignment, we did width sweeps at 153.6M tokens with $n/64$ heads and feedforward width $3.5n$; the default initialization (embedding G/n , readout $G/\sqrt{n}$, hidden-matrix variance $1/k$ for fan-in k); readout multiplier 1. In this problem we consider scaling in **the course baseline parametrization**, not μP . At $n = 512$ this is the default d8 model.

This question changes. The assignment's width sweep is trained with **Hyperball** instead of AdamW, at widths $n \in \{128, 256, 512, 1024\}$ and peak LR $\eta \in \{.006, .01, .015, .022, .03, .045\}$.

(a) (10 points) Fit $\eta^*(n) = c n^p$ to the fitted optima at widths 128, 256, 512 and 1024. Which interval contains p ?

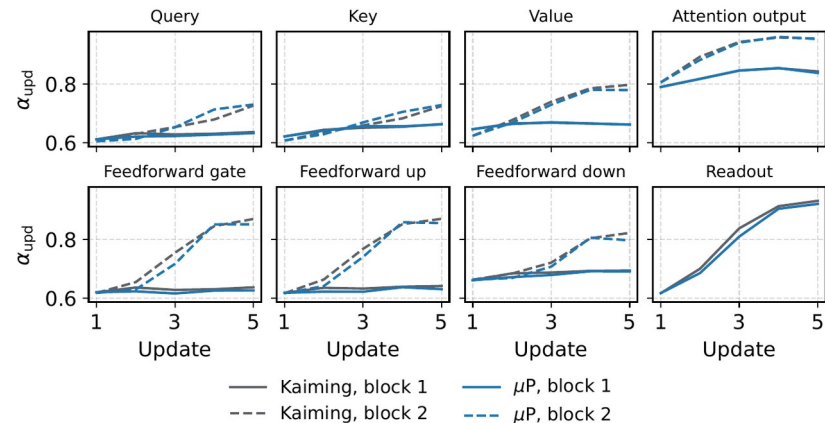
- 1. $(-\infty, -.75)$ · 2. $[-.75, -.25)$ · 3. $[-.25, .25)$ · 4. $[.25, \infty)$

Q6 Assignment data



Solid curves show fitted minima; dashed curves show measured losses at the directly transferred LR.

Width	Recipe	Fitted LR	Fitted loss	Loss at .003
128	Baseline	0.008736	3.616812	3.682442
128	μP	0.002144	3.633573	3.637125
256	Baseline	0.004127	3.382973	3.393821
256	μP	0.001997	3.396404	3.417492
512	Shared	0.002289	3.215923	3.223966



Width 640, at the best sampled LR for each prescription. Hidden-matrix solid and dashed lines identify blocks one and two. The final readout has one curve per prescription. Alignment uses each actual matrix update, its pre-update input, and its own fan-in; the down-projection fan-in is $3.5n$.

Baseline scaling is roughly $1/n$

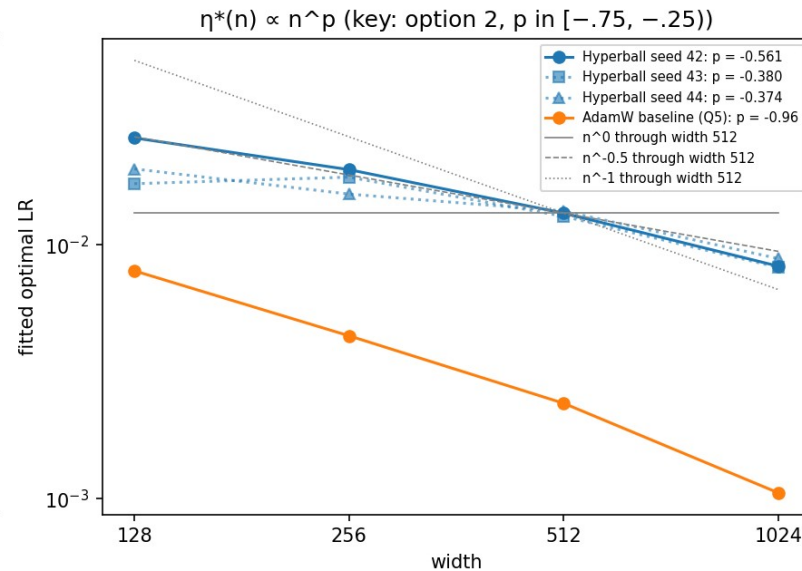
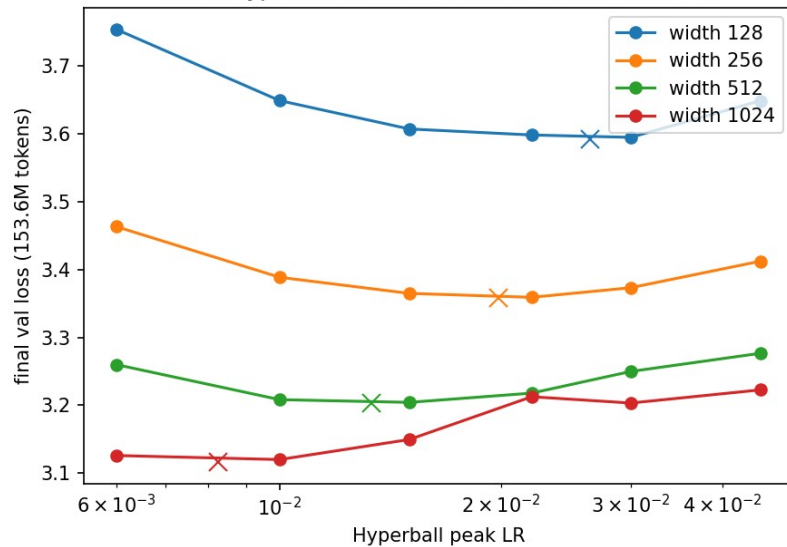
MuP is generally stable, and we know alignment is around 0.75

(a) (10 points) Fit $\eta^*(n) = cn^p$ to the fitted optima at widths 128, 256, 512 and 1024. Which interval contains p ?

1. $(-\infty, -.75)$ 2. $[-.75, -.25)$ 3. $[-.25, .25)$ 4. $[.25, \infty)$

Q6 answer and rationale

Q6(a) Hyperball, course-baseline widths, seed 42

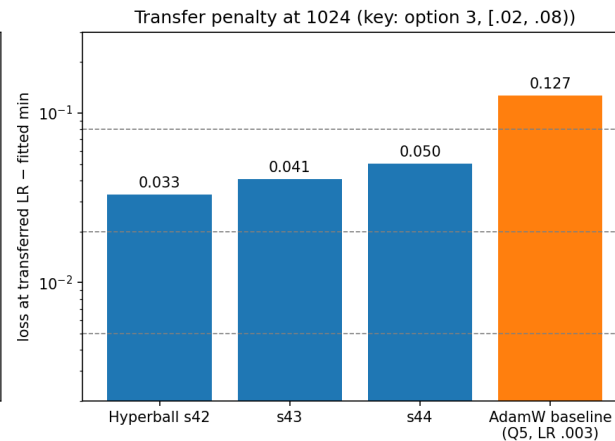
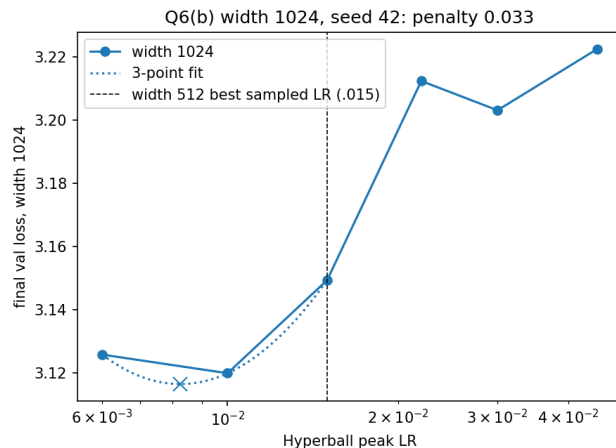


Hyperball takes a step of size $\sim \frac{\eta \|W\|_F}{\|U\|_F} * \|U\|_F = O\left(\frac{1}{\sqrt{m}}\right)$. Base adamW needed $\frac{1}{m}$ scaling in LR, so we expect $\frac{1}{\sqrt{m}}$

Part b

(b) (10 points) At width 1024, which interval contains the transfer penalty of width 512's best sampled LR?

1. $[0, .005)$ 2. $[.005, .02)$ 3. $[.02, .08)$ 4. $[.08, \infty)$



Should expect slightly larger penalty, between adam and mup,
so we expect the mid interval $[0.02, 0.08)$

Q7

This question changes. Batch size $\mathbf{B} = 512$, so training takes eight times fewer updates, with peak LR and WD set by one of five rules:

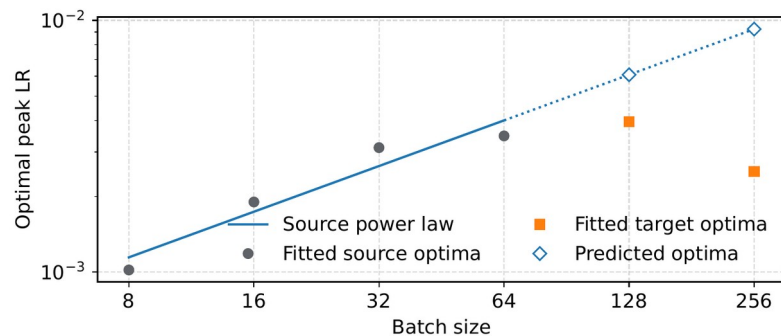
Recipe	Rule	Peak LR η	WD λ	$\eta\lambda (\times 10^{-4})$
A	Keep both	.003	.2	6
B	LR $\times \sqrt{8}$	.00849	.2	17
C	LR $\times 8$	.024	.2	48
D	WD $\times 8$	.003	1.6	48
E	LR $\times \sqrt{8}$ and WD $\times \sqrt{8}$	.00849	.566	48

Let $L_A, \dots, L_E$ denote the losses of recipes A–E.

(a) (10 points) Which recipe gives the lowest loss?

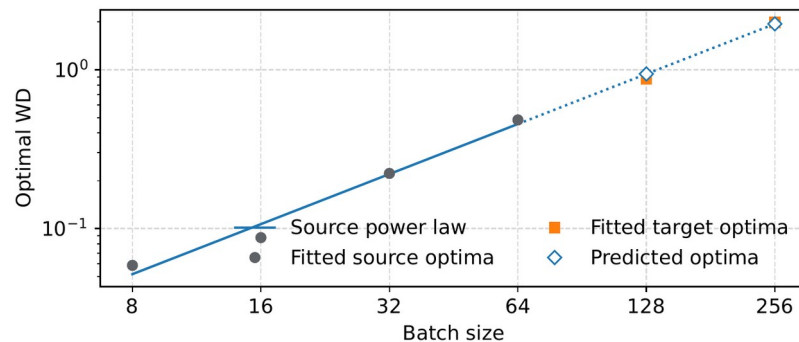
☐ A ☐ B ☐ C ☐ D ☐ E

Assignment data



WD .1. Only batches 8–64 enter the power law. The dotted segment extrapolates to batches 128 and 256; target optima are fitted separately.

B	Predicted LR	Measured loss	Grid best	Difference
128	0.006081	2.965250	2.956287	+0.008964
256	0.009232	3.144123	3.013724	+0.130398



Peak LR .0015. Only batches 8–64 enter the power law. The dotted segment extrapolates to batches 128 and 256; target optima are fitted separately.

B	Predicted WD	Measured loss	Grid best	Difference
128	0.940724	2.939410	2.939511	-0.000101
256	1.943967	2.970108	2.972917	-0.002810

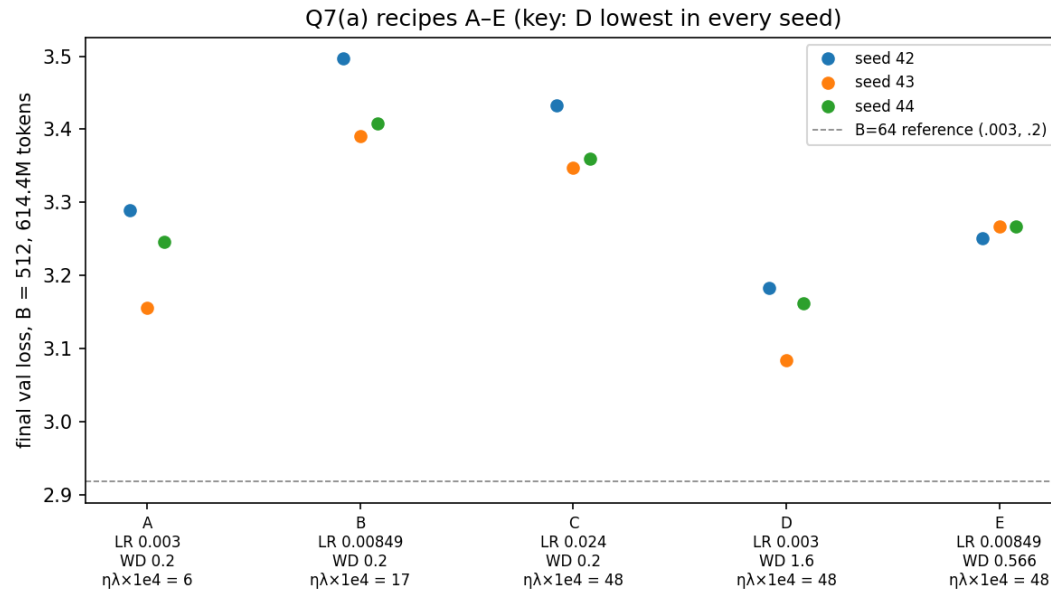
Weight decay scaling is predictable for large batch, LR scaling tapers off

Let $L_A, \dots, L_E$ denote the losses of recipes A–E.

(a) (10 points) Which recipe gives the lowest loss?

A B C D E

Answer and rationale



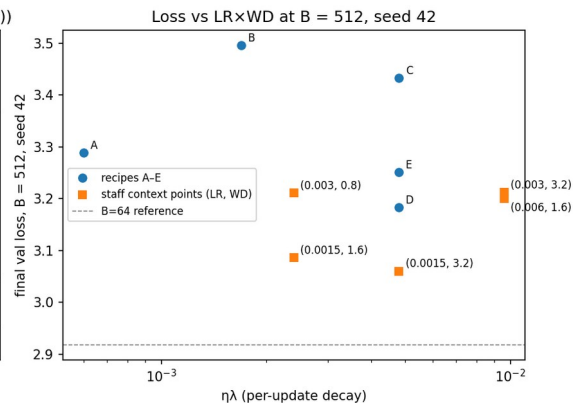
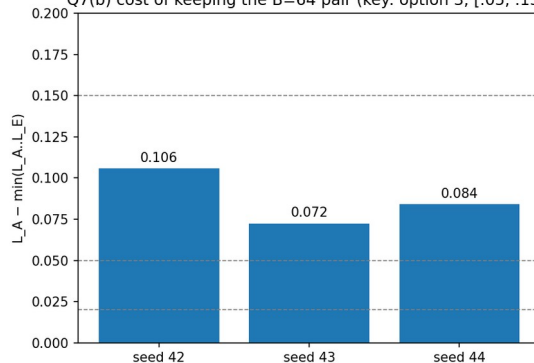
WD scaling turns out to be the “right” thing to do.

Part b answer

(b) (10 points) Which interval contains $L_A - \min(L_A, \dots, L_E)$, the cost of keeping the batch-64 pair?

1. $[0, .02)$ 2. $[.02, .05)$ 3. $[.05, .15)$ 4. $[.15, \infty)$

Q7(b) cost of keeping the B=64 pair (key: option 3, $[.05, .15)$)



We are 2^3 less on the LR*WD product.

We know this should be a decent bit worse, but how much worse?

- One way – 0.15 is much worse! As bad as just naively scaling batch linearly. (0.13)
- We had that each doubling costs ~ 0.01 so $\log_2(8)^2 * 0.01 \approx 0.082$